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Cryptography And Network Security

1.

Given,

Plaintext :- MEETATNOON

key : 7

using ceaser cipher :

Encryption:

$$C = (p + k) \bmod 26$$

0	1	2	3	4	5	6	7	8	9	10	11
A	B	C	D	E	F	G	H	I	J	K	L

12	13	14	15	16	17	18	19	20	21
M	N	O	P	Q	R	S	T	U	V

22	23	24	25
W	X	Y	Z

12	4	4	19	0	19	13	14	14	13
M	E	E	T	A	T	N	O	O	N

For M :- $C = (12 + 7) \bmod 26 = 19 \bmod 26 = 19$ (T)

For E : $C = (4 + 7) \bmod 26 = 11 \bmod 26 = 11$ (L)

For E : $C = (4 + 7) \bmod 26 = 11 \bmod 26 = 11$ (L)

For T : $C = (19 + 7) \bmod 26 = 26 \bmod 26 = 0$ (A)

For A : $C = (0 + 7) \bmod 26 = 7 \bmod 26 = 7$ (H)

For T : $C = (19 + 7) \bmod 26 = 26 \bmod 26 = 0$ (A)

For N : $C = (13 + 7) \bmod 26 = 20 \bmod 26 = 20$ (U)

For O : $C = (14 + 7) \bmod 26 = 21 \bmod 26 = 21$ (V)

For O : $C = (14 + 7) \bmod 26 = 21 \bmod 26 = 21$ (V)

For N : $C = (13 + 7) \bmod 26 = 20 \bmod 26 = 20$ (U)

MEETATNOON ⟶ TLLAHAUVVU

Decryption:

19	11	11	0	7	0	20	21	21	20
T	L	L	A	H	A	U	V	V	U

For decryption

$$P = (C - K) \bmod 26$$

key = 7

For T :- $P = (19-7) \bmod 26 = 12 \bmod 26 = 12$ (M)

L :- $P = (11-7) \bmod 26 = 4 \bmod 26 = 4$ (E)

L :- $P = (11-7) \bmod 26 = 4 \bmod 26 = 4$ (E)

A : $P = (0-7) \bmod 26 = -7 \bmod 26 = 19$ (T)

H : $P = (7-7) \bmod 26 = 0 \bmod 26 = 0$ (A)

A : $P = (0-7) \bmod 26 = -7 \bmod 26 = 19$ (T)

U : $P = (20-7) \bmod 26 = 13 \bmod 26 = 13$ (N)

V : $P = (21-7) \bmod 26 = 14 \bmod 26 = 14$ (O)

V : $P = (21-7) \bmod 26 = 14 \bmod 26 = 14$ (O)

U : $P = (20-7) \bmod 26 = 13 \bmod 26 = 13$ (N)

TLLAHAUVVU ⟶ MEETATNOON

b) Vigenere cipher:

Plain text :- MEETATNOON

key : SON

Encryption:

text :-	12 M	4 E	4 E	19 T	0 A	19 T	13 N	14 O	14 O	13 N
key :-	18 S	20 U	13 N	18 S	20 U	13 N	18 S	20 U	13 N	18 S

$C_i = (P_i + K) \bmod 26$

for M :- $C = (12+18) \bmod 26 = 30 \bmod 26 = 4$ (E)

E :- $C = (4+20) \bmod 26 = 24 \bmod 26 = 24$ (Y)

E :- $C = (4+13) \bmod 26 = 17 \bmod 26 = 17$ (R)

T :- $C = (19+18) \bmod 26 = 37 \bmod 26 = 11$ (L)

A :- $C = (0+20) \bmod 26 = 20 \bmod 26 = 20$ (U)

T :- $C = (19+13) \bmod 26 = 32 \bmod 26 = 6$ (G)

N :- $C = (13+18) \bmod 26 = 31 \bmod 26 = 5$ (F)

O :- $C = (14+20) \bmod 26 = 34 \bmod 26 = 8$ (I)

D :- $C = (14+13) \bmod 26 = 27 \bmod 26 = 1$ (B)

N :- $C = (13+18) \bmod 26 = 31 \bmod 26 = 5$ (F)

MEETATNOON ⟶ EYRLUGFIBF

Decryption:

$P_i = (C_i - K) \bmod 26$

text :-	E (4)	Y (24)	R (17)	L (11)	U (20)	G (6)	F (5)	I (8)	B (1)	F (5)
key :-	S (18)	U (20)	N (13)	S (18)	U (20)	N (13)	S (18)	U (20)	N (13)	S (18)

for E :- $P = (4-18) \bmod 26 = -14 \bmod 26 = 12$ (M)

Y :- $P = (24-20) \bmod 26 = 4 \bmod 26 = 4$ (E)

R :- $P = (17-13) \bmod 26 = 4 \bmod 26 = 4$ (E)

L :- $P = (11-18) \bmod 26 = -7 \bmod 26 = 19$ (T)

U :- $P = (20-20) \bmod 26 = 0 \bmod 26 = 0$ (A)

G :- $P = (6-13) \bmod 26 = -7 \bmod 26 = 19$ (T)

F :- p = (5-18) mod 26 = -13 mod 26 = 13 (N)

I :- P = (8-20) mod 26 = -12 mod 26 = 14 (O)

B :- p = (1-13) mod 26 = -12 mod 26 = 14 (O)

F :- p = (5-18) mod 26 = -13 mod 26 = 13 (N)

EVRLUGFIBF → MEETATNOON

b) Comparsion :-

* In the ceaser cipher the key is fixed for all same element but where as in the vignere cipher if there are any same elment the key value is different.
* In both the ciphers the key value places a major role.
* It the ceaser cipher it uses the 26 Alphabet shift.
* In the vigenere cipher it uses 26^3 shift because of the key value.
* Both the ciphers are used for both encryption and decryption.

c) Vulnerability to cryptanalysis :

* If we use the ceaser cipher it becomes easy to the attacker to know the original text.
* In the ceaser cipher if any two elments are same the the converted cipher text also becomes same which may cause the damage.

Comparing to ceaser cipher vignere cipher will be recommended in the current situation.

Because for the attacker the decoding takes time on the vigenere cipher.

* Compare by both vigenere is good but for the strong encryption both the ciphers doesn't play a good role.
* for the Authentication and Integrity we can go with the AES-256-Gcm.
* which is a modern algorithm with the strong protection.

Given,

Plain text: MATH

$$\text{key (K)} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

Using the Hill cipher :-

Encryption: Plain text :- MATH

12 0 | 19 7

MA | TH

$$C = KP \bmod 26$$

For MA :-

$$= \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 3\times12 + 2\times0 \\ 5\times12 + 7\times0 \end{bmatrix} \bmod 26$$

$$\begin{bmatrix} 36+0 \\ 60+0 \end{bmatrix} \bmod 26$$

$$\begin{bmatrix} 36 \\ 60 \end{bmatrix} \bmod 26$$

$$\begin{bmatrix} 10 \\ 8 \end{bmatrix} \begin{matrix} (K) \\ (I) \end{matrix}$$

MA → KI

For TH :- $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} \bmod 26$

$$= \begin{bmatrix} 3\times19 + 2\times7 \\ 5\times19 + 7\times7 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 57+14 \\ 95+49 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 71 \\ 144 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 19 \\ 14 \end{bmatrix} \begin{matrix} (T) \\ (O) \end{matrix}$$

TH → TO

MATH ⟶ KITO

For Decryption:

Formula: $\boxed{P = K^{-1}(y-b) \bmod 26}$

10 8 19 14
KITO

First we need to find the

$$K^{-1} = \frac{1}{\det K} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$\det(k) = (21-10) = 11$

$11^{-1} \bmod 26 = 1$

$11x \bmod 26 = 1$

$11 \times 19 \bmod 26 = 1$

$\boxed{x = 19}$

$$k^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26$$

$$= 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \bmod 26$$

$$k^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix}$$

10 8 19 14
K I | J O

For :- KI :-

$$\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix} \bmod 26$$

$$\begin{bmatrix} 3 \times 10 + 14 \times 8 \\ 9 \times 10 + 15 \times 8 \end{bmatrix} \bmod 26$$

$$\begin{bmatrix} 30 + 112 \\ 90 + 120 \end{bmatrix} \bmod 26$$

$$\begin{bmatrix} 142 \\ 210 \end{bmatrix} \bmod 26$$

$$\begin{bmatrix} 12 \\ 0 \end{bmatrix} \begin{matrix} (M) \\ (A) \end{matrix}$$

$\boxed{KI \longrightarrow MA}$

For TO :- $\begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix} \begin{bmatrix} 19 \\ 14 \end{bmatrix} \bmod 26$

$\begin{bmatrix} 3\times19 + 14\times14 \\ 9\times19 + 15\times14 \end{bmatrix} \bmod 26$

$\begin{bmatrix} 57+196 \\ 171+210 \end{bmatrix} \bmod 26$

$\begin{bmatrix} 253 \\ 381 \end{bmatrix} \bmod 26$

$\begin{bmatrix} 19 \\ 07 \end{bmatrix} \begin{matrix} (T) \\ LH \end{matrix}$

TO → TH

KITO → MATH

Decryption / encryption

text :- MATH

key :- $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$

Text :- 12 0 | 19 7
MA | TH

encryption :

$C = KXP \bmod 26$

For :- MA $= \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} \bmod 26$

$= \begin{bmatrix} 24+0 \\ 48+0 \end{bmatrix} \bmod 26 = \begin{bmatrix} 24 \\ 22 \end{bmatrix} = \begin{bmatrix} Y \\ W \end{bmatrix}$

For : TH :- $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} \bmod 26$

$$= \begin{bmatrix} 38+21 \\ 76+35 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 59 \\ 111 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 7 \\ 7 \end{bmatrix} \bmod 26 = \begin{bmatrix} 7 \\ 7 \end{bmatrix} \begin{bmatrix} 11 \\ 11 \end{bmatrix}$$

MATH ⟶ VNHH

Decryption:

24 13 7 7
VNHH

$$k^{-1} = \frac{1}{\det k} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \bmod 26$$

$$\det k = (10 - 12)$$
$$= -2$$

Here -2 is not possible to be calculated as we can't find the k^{-1}.

Major Weakness:

* In the Hill cipher we can't find the appropriate Decryption value as there it presents the negative Sign values.
* For finding (k^{-1}) the determinant we can't find if the value of det (k) is (-ve).
* Hence finding the plaintext from cipher text is difficult in the Hill cipher.

3. AES-256-GCM and RSA:

Comparsion:

Scenario:- 200 hospital clinic transfers the (1GB MRI) files to the central server.

Feature	AES-256-GCM	RSA
Type	Symmetric	Asymmetric
key	one shared secret key	public and private key.
Speed	Very fast	Slow process
large files	Suitable	Not Suitable

Key Scalability:-

For 200 clinics Communicating pairwise :-

$$\text{Keys} = \frac{n(n-1)}{2}$$

$$= \frac{200(200-1)}{2} = \frac{200(199)}{2}$$

$$= \frac{39800}{2}$$

$$\boxed{\text{Keys} = 19900}$$

In each of the clinic's there will only communicate with the central Server, 200 shared Aes keys are required.

SA:-

* Here each clinic has only one public-private key pair.
* The Server has one public-private key pair.
* Total = 201 key pairs.

AES is suitable for the large files, whereas RSA is only suitable for the secure key Exchange, not for the larger files.

Hybrid Encryption WorkFlow:-

Hybrid encryption combines both the AES and RSA

Sending towards the clinic:-

* It is validate the Server's digital certificates.
* Generate a random 256-bit DES
* It generates a unique of the both AES and RSA.

Receiving Servers.

* It will be generating only a two where in the AES it is suitable for the file exchanging.
* For every time there will be authentication where it provides the more Security.
* It will be accepted only if the authentication is correct else not.

Secure key Storage:

* It uses the audit logs

* It is used mainly for the authentication.
* It will be used of the MFA for storing the keys.
* It will be played as role based access.
* It always main a database for the new users.

4. RSA demonstration:-

Given, prime number :- $p = 11$

$q = 19$

$e = 7$

To find $\boxed{n = p \times q}$

$n = 11 \times 19$

$\boxed{n = 209}$

To find Euler's totient ϕn

$\phi n = (p-1) \times (q-1)$

$\phi n = (11-1) \times (19-1)$

$\phi n = 10 \times 18$

$\boxed{\phi n = 180}$

The values of $n = 209$

$\phi n = 180$

gcd $(e, \phi n) = 1$

$(7, 180) = 1$

To find the encrypted d:-

$7 \times d \equiv 1 \pmod{\phi n}$

$7d \equiv 1 \bmod 180$

$7 \times 103 = 1 \bmod 180$

7×24 = 168
7×25 = 175
182 mod 7

$c = M^e \bmod n$ $\quad$ $d = 103$

Here exponent $e = 7$

Now finding public key & private key.

Encryption (Sender)

$M = 5$

$C = M^e \bmod n$

$c = 5^7 \bmod 209$

$c = 78125 \bmod 209$

$c = 168$

Decryption (Receiver)

$M = c^d \bmod n$

$M = 168^{103} \bmod 209$

$M = 17304 \times 10^{-5} \bmod 209$

$M = 5$

Why RSA is needed:-

* Modern RSA uses 2048 bit or 3072 bit modulo.
* Small keys can be broken through the factorization attacks.

* It is vulnerable to ciphertext manipulation
* Text book is deterministic

Padding Method:

* RSA - OEAP - Secure encryption
* RSA - PSS - Secure digital Signatures.

5. Given,

Plaintext: INSTRUMENTS

key = 3

caeser cipher:

Encryption Formula:-

$C = (P + k) \bmod 26$

Decryption

$C = (P - k) \bmod 26$

8 13 18 19 17 20 12 4 13 19 18
INSTRUMENTS

For I = (8+3) mod 26
= 11 mod 26 = 11 (L)

N = (13+3) mod 26
= 16 mod 26 = 16 (Q)

S = (18+3) mod 26
= 21 mod 26 (V)

T = (19+3) mod 26
= 22 mod 26 (W)

R = (17+3) mod 26
= 20 (U)

U = (20+3) mod 26
= 23 mod 26 (X)

M = (12+3) mod 26
= 15 mod 26 (P)

E = (4+3) mod 26
= 7 mod 26 (H)

N = (13+3) mod 26
= 16 mod 26 (Q)

T = (19+3) mod 26 = 22 (W)

S = (18+3) mod 26 = 21 (V)

INSTRUMENTS → LQVWUXPHQWV